TCOIMBATORE INSTITUTE OF TECHNOLOGY

(GOVERNMENT AIDED AUTONOMOUS INSTITUTION)

CIVIL AERODROME POST, COIMBATORE-641014.

PLACEMENT CELL

PLACEMENT YEAR 2023 - 2024

NAME : Tamilarasan S

ROLL NUMBER : 2003094

DEPARTMENT : Electrical and Electronics Engineering

EMAIL ID : tamilarasan20225@gmail.com

CONTACT NUMBER : 6369545851

COMPANY NAME : ZOHO

COMPANY TYPE : NON-DREAM

JOB DESIGNATION : MEMBER TECHNICAL STAFF (SDE)

SALARY (CTC) : 5,60,000

INTERN OFFERED ? : Yes

BOND : No

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Quiz(Offline)

Round details: General and Technical Aptitude

Areas covered: Trains, Time, speed &distance, Probability, Percentage, Averages and C Program output finding.

Your Comments: Constant practice in Aptitude and strong understanding in C fundamentals would be helpful to clear this round.

Have you cleared the round : Yes

Details about Questions on 1st round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)

ROUND 2 :

Platform: Any IDE of our choice (OFFLINE)

Round details: Problem Solving

Areas covered: Linked List, Arrays, Bit Manipulation, String

Your Comments: Constant practice in problem solving would be really helpful. We were divided into multiple groups and each group is assigned with a mentor once we solve a problem the solution should be explained to the mentor and they will check the code by providing edge cases and once all the edge cases are passed you are allowed to solve the next question. Make sure your code passes all the edge cases before calling the mentor (3hrs).

Have you cleared the round : Yes

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Programming	**Given a value n representing no of friends and a value m. n friends are standing in a circle and on each iteration we have to go through m friends and the mth friend gets eliminated and this cycle continues till there is one friend left.** **Example:** **INPUT:** **n=5,m=6** **OUTPUT:** **4**	**Created a queue and inserted n friends inside a queue.Then for each m-1 iteration pop and push the friend .At m'th iteration pop the friend.** **Continue this until only one friend remains.**	https://practice.geeksforgeeks.org/problems/josephus-problem/1
	Valid Substring: Given a string of parentheses, the task is to find the valid substring, a valid substring, for every open parentheses in the substring it should a have a closing parentheses. **Input:** **Str=" ()((((((()))"** **Output: 6**	**Used a stack for keeping track of the open parentheses and whenever a close parentheses is encountered and if the stack has a open parentheses at the top it gets popped and the count is increased**	https://www.geeksforgeeks.org/length-of-the-longest-valid-substring/
	Given an array of numbers and the largest number should be made out from the array by concatenation and it can be concatenated in any order such that the largest number should be made out of it. **Input: 9 3 30 56 6** **Output: 9656330**	**Sorted the array with a modification in the comparator. While comparing two numbers for sorting, consider a and b as two numbers it is joined as 'ab' & 'ba' if 'ab' is greater than 'ba' it won't be sorted else it will be sorted.**	https://www.geeksforgeeks.org/given-an-array-of-numbers-arrange-the-numbers-to-form-the-biggest-number/
	Given 8 32-bit integers form a new number such that the 1 st 4bits of the new number is the last 4 bits(32-29) of the 1 element and next 4 bits of the new number is 2nd last 4bits of the 2nd integer(28-25) and this follows to form 32 bits for the new number. **Input:**	**Used bitwise and operator for extracting the suitable bits for each integer using the right shift operator and used a normal binary to decimal conversion.**	

	8 **1 2 3 4 5 6 7 8** **Output:** **8**		
	Given an array of integers find the length of the longest mountains that are present in the array. Mountain represents a subarray that has values increasing followed by decreasing values. **Input:** **5 0 1 0 1 0** **Output:** **2**	**Whenever increase values are noted in array checked if it is a mountain. If it is a mountain then updated the answer for maximum value**	**https://www.geeksforgeeks.org/longest-mountain-subarray/**

ROUND 3 :

Platform: Any IDE of our choice (OFFLINE)

Round details: Design Round

Areas covered: OOPS

Your Comments: Have a good understanding of oops concepts such that it will be easy for you to implement the given tasks (3hrs) use a Object Oriented language and I used CPP.

Have you cleared the round : Yes

Details about Questions on 3rdround:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Long coding	**We were asked to design a railway ticket booking system and multiple subtasks were given. 1. Book tickets. 2. Cancel tickets. 3. Give the profit on particular date 4. Give the ticket details on a given date**	**Implemented OOPS concepts for implementing different modules for subtasks.**	

ROUND 4 :

Platform: OFFLINE

Round details: Technical Interview 1

Areas covered: Problem Solving, DSA, Computer Networks, DBMS, OOPS, Operating System, Puzzles, Project.

Your Comments: This round went for around 1hr45min and the interviewer was keen on my approach to problem. Be thorough with whatever you have mentioned in your resume. Explain everything in a detailed way.

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Computer Networks	**Explain the OSI model and explain how internet works. Explain https request.**	**Explained each layer of osi models with examples in brief manner. The interviewer also asked about protocols involved in each layer and data format for those protocols. I explained the data format for some protocols. Then explained how https requests works while writing some code.**	
DSA	**Explain Red black trees, B+ trees. Explain how avl tree rotation takes place. Infix, prefix, postfix conversions.**	**Explained the concepts in a brief manner with example.**	
DBMS	**What is triggers in database. Normalization in dbms. What is ER model. Explain candidate key, primary key etc.. ACID properties. Crud operation**		
Operating System	**Explain threads, Semaphore. How will you solve concurrency problem in os. Memory allocation.**		

	What Is virtual memory.		
Puzzles	**1. In Gold Bar Puzzle, You've someone working for seven days and a gold bar to pay him. The gold bar is segmented into seven connected pieces. You must give them a piece of gold at the end of every day. What and where are the fewest number of cuts to the bar of gold that will allow you to pay him 1/7th each day?** **2. Given two hourglass of 4 minutes and 7 minutes, the task is to measure 9 minutes.**		https://prepinsta.com/top-100-puzzles/gold-bar-puzzle/ https://www.geeksforgeeks.org/puzzle-27-hourglasses-puzzle/
Project	**Explain your project ?** **Why you have chosen this project.** **Interviewer also asked some general questions**		

ROUND 5 :

Platform: OFFLINE

Round details: General HR

Areas covered:

Your Comments This round went for around 15min and only general questions were asked like family background and feedbacks from previous technical rounds

Have you cleared the round : Yes

AREAS TO PREPARE :

1. Be strong in problem solving and programming concepts.
2. Be strong in aptitude
3. Be strong in DSA, Computer Networks, DBMS and OOPS

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

1.APTITUDE:
https://youtube.com/playlist?list=PLpyc33gOcbVA4qXMoQ5vmhefTruk5t9lt&si=x_8bu2ToCveW6mJ0
2.DBMS: https://www.javatpoint.com/dbms-tutorial
3.COMPUTER NETWORKS: https://www.geeksforgeeks.org/computer-network-tutorials/
4.Problem Solving:
(i) https://www.geeksforgeeks.org/explore/
(ii)https://leetcode.com/problemset/all/
(iii)https://takeuforward.org/interviews/strivers-sde-sheet-top-coding-interview-problems/
5.OOPS: https://www.geeksforgeeks.org/object-oriented-programming-in-cpp

OVERALL EXPERIENCE : Good

GENERAL TIPS :

- Explain your method of approach to the mentors before coding in the third Round.
- Try to optimise your solution
- Clarify your doubts before starting to code.
- Have a calm approach to every problem.

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PLACEMENT CELL

PLACEMENT YEAR 2023 - 2024

NAME : DEEPAK G

ROLL NUMBER : 2004006

DEPARTMENT : ELECTRONICS AND COMMUNICATION ENGINEERING

EMAIL ID : deepakgowthaman2203@gmail.com

CONTACT NUMBER : 9176803725

COMPANY NAME : ZOHO

COMPANY TYPE : NON-DREAM

JOB DESIGNATION : MEMBER TECHNICAL STAFF (SDE)

SALARY (CTC) : 5,60,000

INTERN OFFERED ? : YES

BOND : NO

*(Please comment this section **in detail** to guide your juniors in a bright way)*

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Quiz (OFFLINE)

Round details: General and Technical Aptitude

Areas covered: Trains, Time, speed &distance, Probability, Percentage, Averages and C Program output finding.

Your Comments: Constant practice in Aptitude and strong understanding in C fundamentals would be helpful to clear this round.

Have you cleared the round : YES

Details about Questions on 1st round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)

ROUND 2 :

Platform: Any IDE of our choice (OFFLINE)

Round details: Problem Solving

Areas covered: Linked List, Arrays, Bit Manipulation, String

Your Comments: Constant practice in problem solving would be really helpful. We were divided into multiple groups and each group is assigned with a mentor once we solve a problem the solution should be explained to the mentor and they will check the code by providing edge cases and once all the edge cases are passed you are allowed to solve the next question. Make sure your code passes all the edge cases before calling the mentor (3hrs).

Have you cleared the round : YES

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Programming	Given a value n representing no of friends and a value m. n friends are standing in a circle and on each iteration we have to go through m friends and the mth friend gets eliminated and this cycle continues till there is one friend left. Example: INPUT: n=5,m=6 OUTPUT: 4	Created a circular Linked List for n friends and on each iteration ,travelled through m friends and removed one by one till there is one friend left	https://practice.geeksforgeeks.org/problems/josephus-problem/0
Programming	Valid Substring: Given a string of parentheses, the task is to find the valid substring, a valid substring, for every open parentheses in the substring it should a have a closing parentheses. Input: Str=" ()((((()))" Output: 6	Used a stack for keeping track of the open parentheses and whenever a close parentheses is encountered and if the stack has a open parentheses at the top it gets popped and the count is increased	https://www.geeksforgeeks.org/length-of-the-longest-valid-substring/
Programming	Given an array of numbers and the largest number should be made out from the array by concatenation and it can be concatenated in any order such that the largest number should be made out of it. Input: 9 3 30 56 6 Output: 9656330	Sorted the array with a modification in the comparator. While comparing two numbers for sorting, consider a and b as two numbers it is joined as 'ab' & 'ba' if 'ab' is greater than 'ba' it won't be sorted else it will be sorted.	https://www.geeksforgeeks.org/given-an-array-of-numbers-arrange-the-numbers-to-form-the-biggest-number/
Programming	Given 8 32-bit integers form a new number such that the 1st 4bits of the new number is the last 4 bits(32-29) of the 1 element and next 4 bits of the new number is 2nd last 4bits of the 2nd integer(28-25) and this follows to form 32 bits for the new number. Input: 8	Used bitwise operators for extracting the suitable bits for each integer using the right shift operator and used a normal binary to decimal conversion.	

	1 2 3 4 5 6 7 8 **Output:** **8**		
Programming	**Given an array of integers find the number of mountains that are present in the array. Mountain represents a subarray that has values increasing followed by decreasing values.** **Input:** **5** **0 1 0 1 0** **Output:** **2**	**I iterated over the array and used a variable to indicate whenever there was an increase in value. I kept track of the mountain and incremented the count whenever a decreasing value was encountered.**	

ROUND 3 :

Platform: Any IDE of our choice (OFFLINE)

Round details: Design Round.

Areas covered: OOPS

Your Comments: Have a good understanding of oops concepts such that it will be easy for you to implement the given tasks (3hrs) use a Object Oriented language and I used CPP.

Have you cleared the round : YES

Details about Questions on 3rdround:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
LONG CODING	**We were asked to design a railway ticket booking system and multiple subtasks were given.** **1. Book tickets.** **2. Cancel tickets.** **3. Give the profit on particular date** **4. Give the ticket details on a given date.**	**Implemented OOPS concepts for implementing different modules for subtasks.**	

ROUND 4 :

Platform: OFFLINE

Round details: Technical Interview 1

Areas covered: Problem Solving, DSA, Computer Networks, DBMS, OOPS, Machine Learning.

Your Comments: This round went for around 1hr45min and the interviewer was keen on my approach to problem. Be thorough with whatever you have mentioned in your resume. Explain everything in a detailed way.

Have you cleared the round : YES

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Problem Solving	**BLOOM FILTER:** **Given a filter which uses hashing function and gives user the data about, whether a number exists in the set or not and the filter uses a bitwise array for giving the details whether the number is present or not.**	**As I can't remember the syntax for implementing bitwise array, I gave the interviewer with an alternate solution of implementing the same with integers by using them as bitwise arrays and implemented the logic for finding whether the number is present or not using the hashing function.**	
Computer Networks	**Explain how the Internet works.**	**Explained the concept in a detailed way with respect to the OSI layers and while explaining the interviewer asked questions on how each protocol works and I also explained the protocols also.**	
DBMS	**ACID properties.**	**Explained the ACID properties of DBMS with examples.**	
DBMS	**Concurrency control**	**Explained how concurrency is achieved in Database using Locks and Isolation.**	

SQL	**Given tables and asked to query on it and also asked to explain the flow of execution (one easy query and one complex query).**	**Used joins for querying the table and explained the flow of execution.**	
DSA	**Given different scenarios and asked to give the appropriate data structures that I will use in those situations.**	**Analyzed each scenario and after analyzing the needs gave the DS that satisfies all those needs**	
Project	**Asked to explain my current project and asked multiple questions on the ML algorithms used in the project.**	**Explained the project in detail.**	

ROUND 5 :

Platform: OFFLINE

Round details: Technical Interview 2

Areas covered: CPP,C, OOPS, Machine Learning.

Your Comments: This round went for around 45min and was mainly focused on OOPS and ML.

Have you cleared the round : YES

Details about Questions on 5thround:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
C	**Asked about dynamic memory allocation and the internal working of it**	**Explained Dynamic memory allocation in a detailed way (malloc, calloc, realloc).**	
OOPS	**Asked to explain OOPS in detail**	**Explained OOPS in detail.**	
OOPS	**Asked to write code for overriding and asked to give an overloading example with respect to mobile phone.**	**Wrote a CPP code for overriding with virtual functions and explained the internal working of virtual functions**	

		and virtual table and also gave a real time example for overloading with respect mobile phone (Normal call and Internet call).	
Machine Learning	**Asked to explain the ML algorithms used in the project and asked questions on it.**	**Explained the ML algorithms used in project.**	

ROUND 6 :

Platform: OFFLINE

Round details: GENERAL HR

Areas covered:

Your Comments: This round went for around 15min and only general questions were asked like family background and feedbacks from previous technical rounds.

Have you cleared the round : YES

Details about Questions on 6th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)

AREAS TO PREPARE :

1. **Be strong in problem solving and programming concepts.**
2. **Be strong in aptitude**
3. **Be strong in DSA, Computer Networks, DBMS and OOPS.**

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :
1.APTITUDE:https://youtube.com/playlist?list=PLpyc33gOcbVA4qXMoQ5vmhefTruk5t9lt&si=x_8bu2ToCveW6mJ0
2.DBMS: https://www.javatpoint.com/dbms-tutorial
3.COMPUTER NETWORKS: https://www.geeksforgeeks.org/computer-network-tutorials/
4.Problem Solving:
(i) https://www.geeksforgeeks.org/explore?page=1&sortBy=submissions
(ii)https://leetcode.com/problemset/all/
(iii)https://takeuforward.org/interviews/strivers-sde-sheet-top-coding-interview-problems/
5.OOPS: https://www.geeksforgeeks.org/object-oriented-programming-in-cpp/

OVERALL EXPERIENCE : It was a good experience. The HRs were so friendly.

GENERAL TIPS : Have a constant practice on Problem Solving and aptitude.

NEVER LOOSE YOUR HOPE